DRONE(UAV)

TECHNOLOGY STUDENT ASSOCIATION

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2216003

Table of Contents

Project Overview

This record captures the design, manufacture and testing of two custom quadcopter drones to the 2026 TSA Safari Rescue competition. The challenge makes teams drive UAVs through a maze, find simulated animals and pick them using a claw attached to the airplane.

The drones were constructed by our team using raw materials. Each of the parts is chosen according to purpose: low weight, high thrust, and control. These drones use T-Motor F7 flight controllers, Betaflight firmware and Emax ECO II 2207 brushless motors on TBS Source One V5 carbon fiber frames.

Both drones are configured identically both in hardware and firmware. This makes maintenance easier, minimizes stocking of spare parts and lets pilots change airplanes without having to relearn the controls. Unless indicated in the documentation, both units are covered by documentation.

Technical Specifications

The table below summarizes the core specifications for each drone. Both drones use identical hardware.

Frame	TBS Source One V5, carbon fiber, 5-inch class
Flight Controller	T-Motor F7 (STM32F722RET6, MPU-6000 gyro)
ESC	T-Motor F55A PRO II 4-in-1, bidirectional DShot300
Motors	EMAX ECO II 2207 (x4 per drone)
Battery	Lumenier N2O 1300mAh 6S 150C LiPo
Firmware	Betaflight 4.4 (upgrading to 6.0 beta)
Receiver Protocol	Serial (via UART), SBUS
Propellers	5-inch tri-blade (blue)
Landing Gear	3D printed A-frame, 5.4 inches clearance
Claw	Servo-actuated 3-finger gripper, AUX 3 channel
Weight (no battery)	Approx. 410g (with claw and landing gear)
Weight (with battery)	Approx. 620g
Payload Capacity	Approx. 150g
Flight Time (loaded)	Approx. 5-6 minutes

Flight Controller Details

It is powered by the T-Motor F7 flight controller with STM32F722RET6 processor and MPU-6000 gyroscope. it will take LiPo input (12V to 25.2 V) 3-6S and supply 5 V/2 A on the receiver side and 10 V/2 A on the video transmitters side. The board has a size of 37mm x 37mm and a 30.5mm x 30.5mm mounting pattern with a weight of 8.4g. The telemetry and SBUS are supported on all UART ports thus the receiver does not need an individual UART assignment.

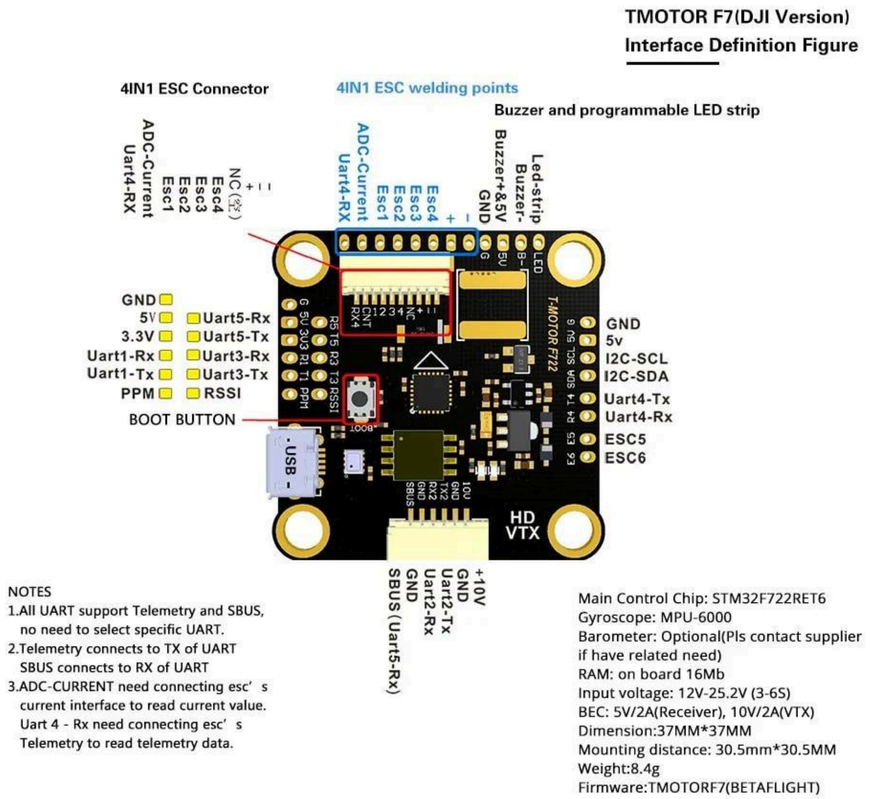


Figure 1: T-Motor F7 Flight Controller Interface Diagram

ESC Configuration

The T-Motor F55A PRO II 4-in-1 ESC has four motor outputs on one board. It is compatible with bidirectional DShot300 that allows RPM telemetry of dynamic notch filtering in Betaflight. The output of each ESC has a continuous rating of up to 55A, which is plenty to allow the EMAX 2207 motors. Motor protocol is configured to DSHOT300 within the Betaflight Motors tab, with bidirectional DShot turned on to filter the RPM.

Motor Selection

We selected EMAX ECO II 2207 because it is priced, weighed and performed well. These motors generate high thrust at 5-inch propellers at moderate current. Its 2207 size stator size has more torque than smaller 1806 or 2004 class motors, making it more responsive to execute precision maneuvers needed to retrieve animals. The direction of the motors is set to Quad X structure where motor 1 and 4 rotate clockwise, motor 2 and 3 rotate counterclockwise.

Build Log

Step 1: Frame Assembly

We started with the TBS Source One V5 carbon fiber frame. This frame is shipped in the form of a package of flat carbon fiber plates which are bolted. The bottom plate is the primary structural component and the four arms are projected outward in approximately 45 degrees. Each arm was firmened to the center stack using M3 screws and nylon lock nuts to ensure that it does not loosen up due to vibrations.
It has a two-plate sandwich construction. The flight controller stack and ESC are on the bottom plate. Its upper plate clamps it all together with aluminum standoffs forming a hardened case that shields internal electronics. The height of the standoff has sufficient space to allow the FC stack, receiver, and all the wiring.

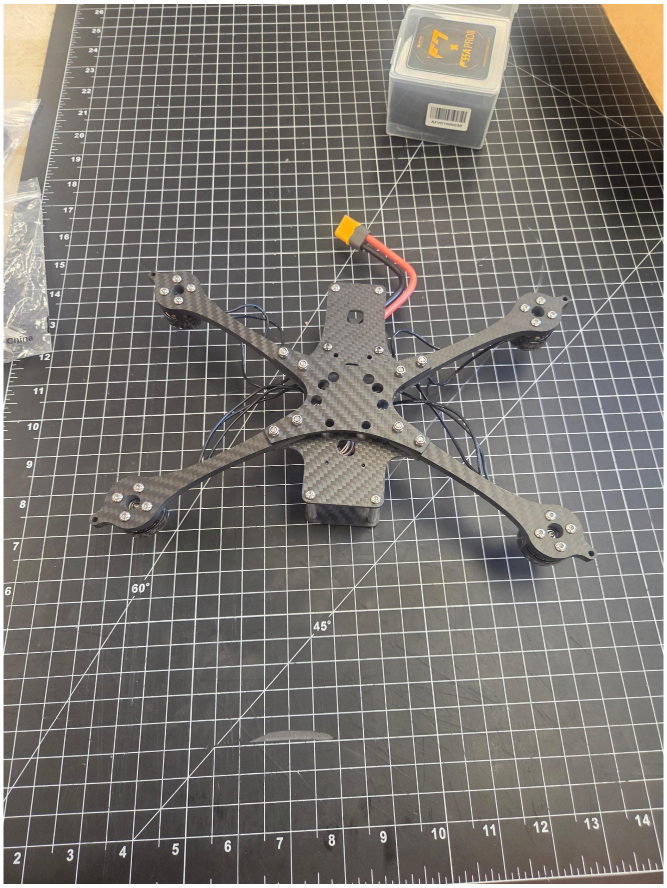


Figure 2: TBS Source One V5 frame, top view, with motors mounted

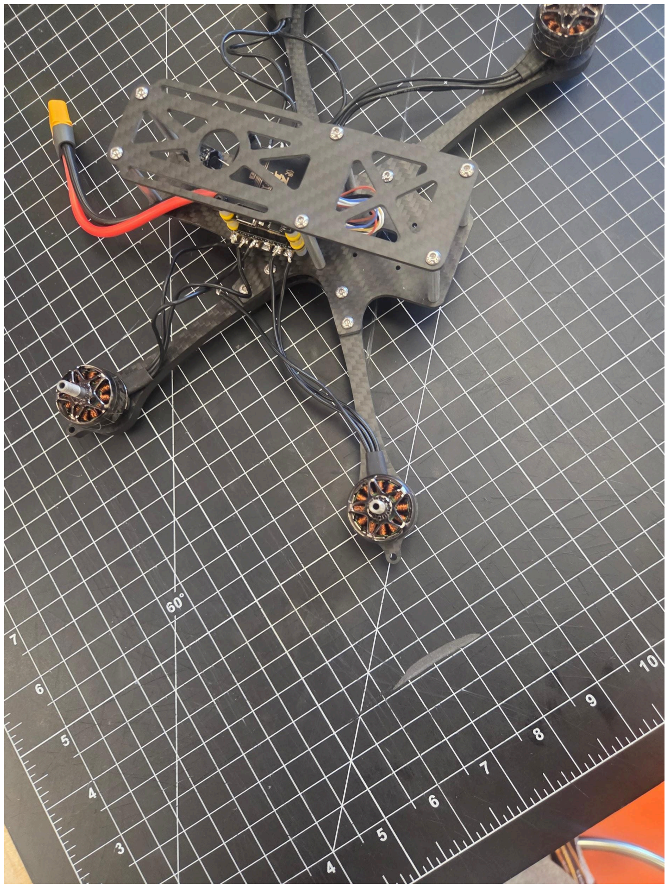


Figure 3: Underside of the frame showing motor wire routing and ESC connections

Step 2: Electronics Installation

To install the T-Motor F55A PRO II ESC, we screwed it to the bottom plate, with soft-mount grommets to reduce vibration. It is mounted directly under the T-Motor F7 flight controller which is directly connected via a standardized 4-in-1 ESC connector. This stacking design maintains all high-current wiring short and centred, minimising electromagnetic interference.

The receiver screws on the top plate with foam tape on the back. Its antennas are not attached to the carbon fiber in order to prevent a decrease in the signal. All the signal wires in the frame were channeled through the cable routes to avoid being in contact with propellers.

Figure 4: Left side view showing EMAX motors, receiver antennas, and internal wiring

Step 3: Soldering

The ESC pads were soldered to all the motor leads. We tinned both the ESC pads and the motor wires and then joined them together and finally added solder at 380 degrees Celsius to make clean solder joints. All the connections were checked under magnification to ensure complete wetting and absence of cold joints.
A battery lead is an XT60 connector that is soldered onto the pads of the ESC power input. To further insulate the XT60 solder cups, we placed heat shrink tubing on all uncovered joints and used a liquid electrical adhesive on the solder cups.

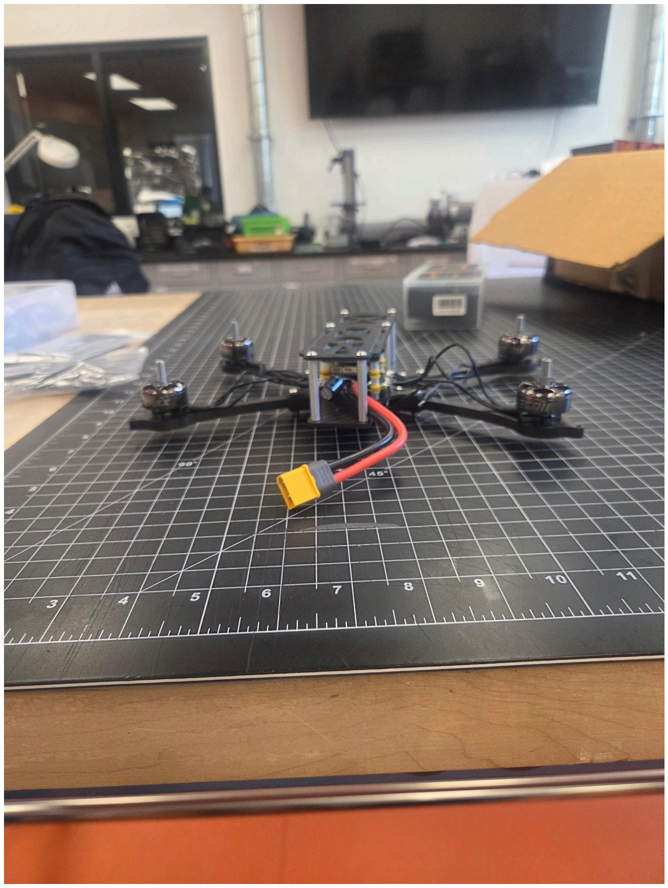


Figure 5: Rear view showing XT60 battery connector and standoff assembly

Step 4: Motor and Propeller Installation

The arm tips are screwed to each EMAX ECO II 2207 motor with four M3 screws. The direction of motor rotation switches (clockwise and counterclockwise) among the diagonal pairs in order to equalize torque. We checked proper rotation in Betaflight motor test mode and then proceeded to fit propellers.
Propellers are 5 inch tri- blade (blue). We tried different pitch settings and found the one that balances thrust and efficiency at our current draw level of 6S battery our 6S battery is capable of.

Figure 6: Front view showing four brushless motors and top plate assembly

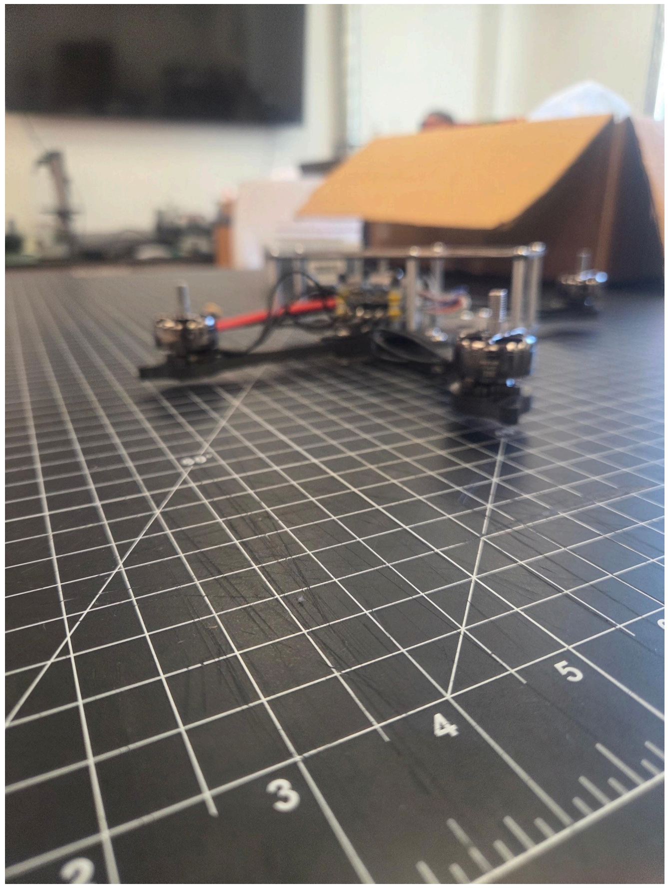


Figure 7: Side profile of the completed drone

Landing Gear

This drone must have its claw mechanism cleared on the ground to ensure that the ground is clear during takeoff and landing. An A-frame landing gear is designed which fits on the bottom of the drone frame.
The 3D printed landing gear is made in PLA, 5.4 inches high. A-shape gives a broad, solid base and reduces the amount of weight and material used. Crossbar of A reinforces rigidity and does not allow splaying of the legs under load. Frames The frame is fitted with two landing gear units so that it has four points of contact with the ground.
This design was selected instead of the conventional skid-type landing gear because in this design the claw can be conveniently suspended between the legs in its A-frame form. This implies that the drone is able to land with a payload already in its grip without the payload coming in touch with the ground.

Figure 8: 3D printed A-frame landing gear component

Claw Mechanism

The Safari Rescue challenge is to retrieve simulated animals in the course. Our gripper is a three-finger servo-actuated gripper that is attached to the bottom of the drone frame.
The claw employs a mechanism with gears. One servo motor operates a gear in the middle one of which opens and closes all three fingers. Three-finger grip offers a secure, centralized grip over two-finger options. The servo is fed by the AUX channel 3 of a flight controller which is set to a switch in the transmitter.
The gripper can hold products of about 60mm in diameter. We experimented with foam animal figures of weight ranging between 20g and 80g. All test objects had adequate grip force with the full range of travel of the servo.

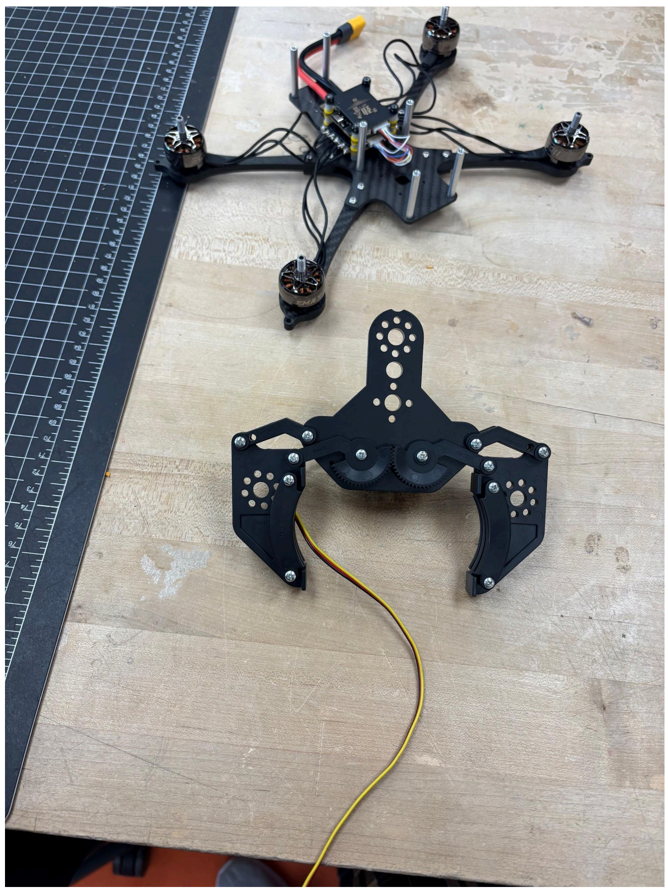


Figure 9: Three-finger gripper mechanism with servo wiring, shown next to the drone frame

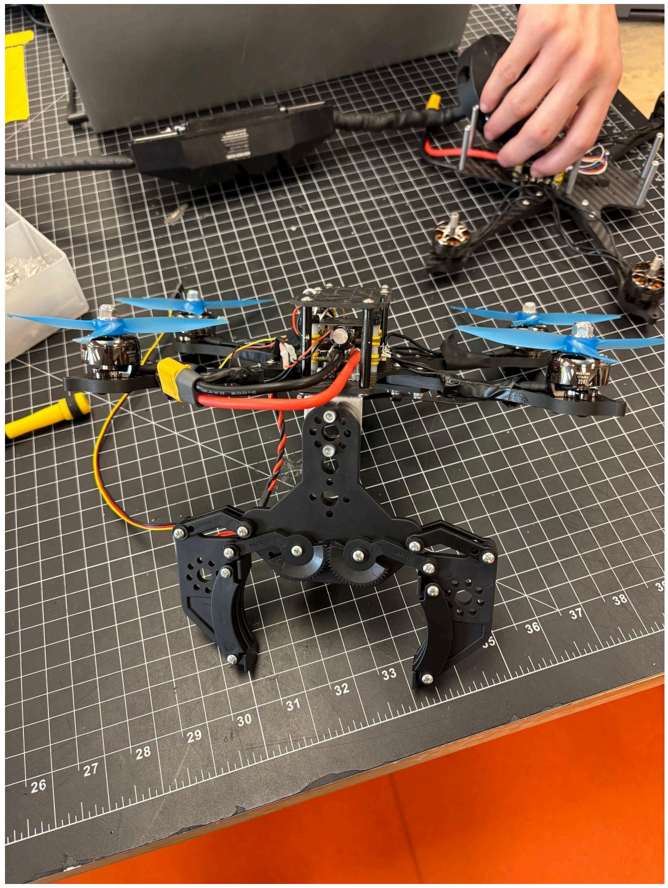


Figure 10: Claw mounted to the drone with blue propellers installed, ready for testing

Firmware and Configuration

Both drones use Betaflight 4.4 targeting the TMOTORF7 firmware, and will be upgraded to Betaflight 6.0 beta once altitude hold has stabilized. The entire setup was done via the Betaflight Configurator. Our precise settings are reported in the screenshots below.

Configuration Tab

The gyro update rate will be 8.00 kHz and the PID loop rate will be 4.00 kHz. The angle mode support is set to the accelerator. Board alignment will be CW 0 degrees and GYRO/ACCEL will be the first gyro. DShot Beacon Configuration receives the tone 1 with making RX_LOST enabled to find the drone in case of signal loss.

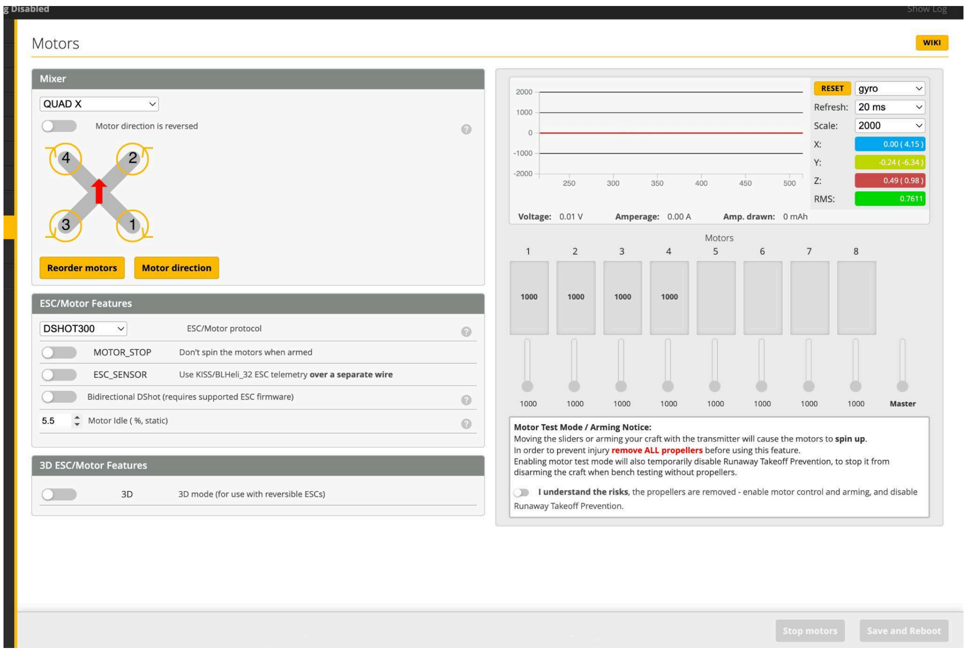


Figure 11: Betaflight Configuration Tab

Motors Tab

Motor protocol DSHOT300 Quad X. DShot is bidirectional, allowing RPM-based filtering. The motor idle value will be 5.5. The motor arrangement is Betaflight-conventional, with motor 1 (back right), and motor 2 (front right), and motor 3 (back left), and motor 4 (front left). Directions of the motors do not reverse.

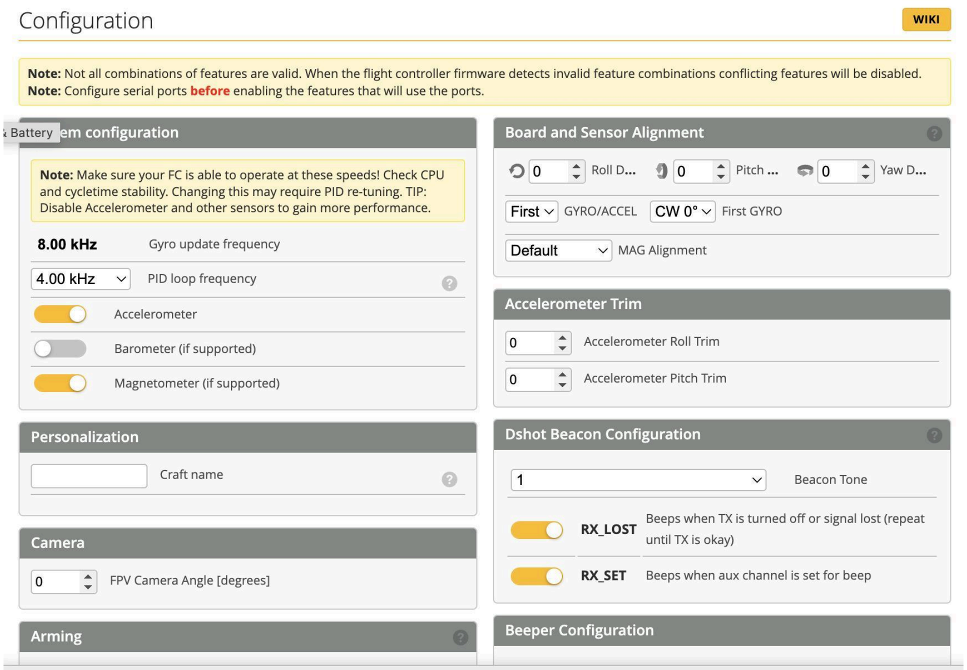


Figure 12: Betaflight Motors Tab

Modes Tab

Three modes are configured. ARM is assigned to AUX 1. ANGLE mode is assigned to AUX 2, providing self-leveling for less experienced pilots during practice. CAMERA CONTROL 1 is assigned to AUX 3, which controls the claw servo.

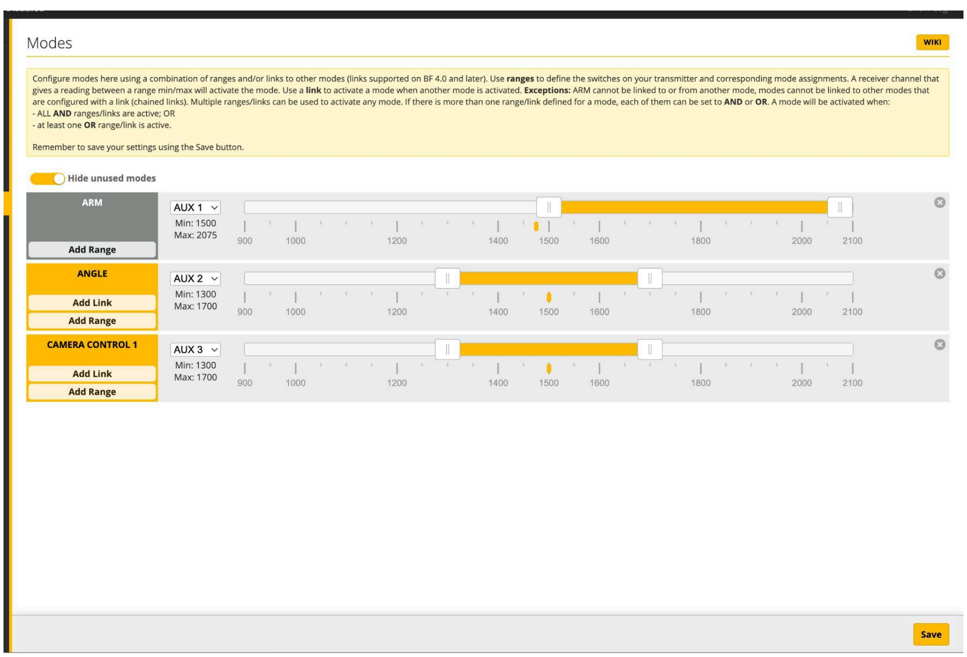


Figure 13: Betaflight Modes Tab

Ports Tab

UART2 has Serial Rx enabled for the receiver connection. UART1 has VTX (TBS SmartAudio) assigned under Peripherals for video transmitter control. All other UARTs remain at default settings with 115200 baud.

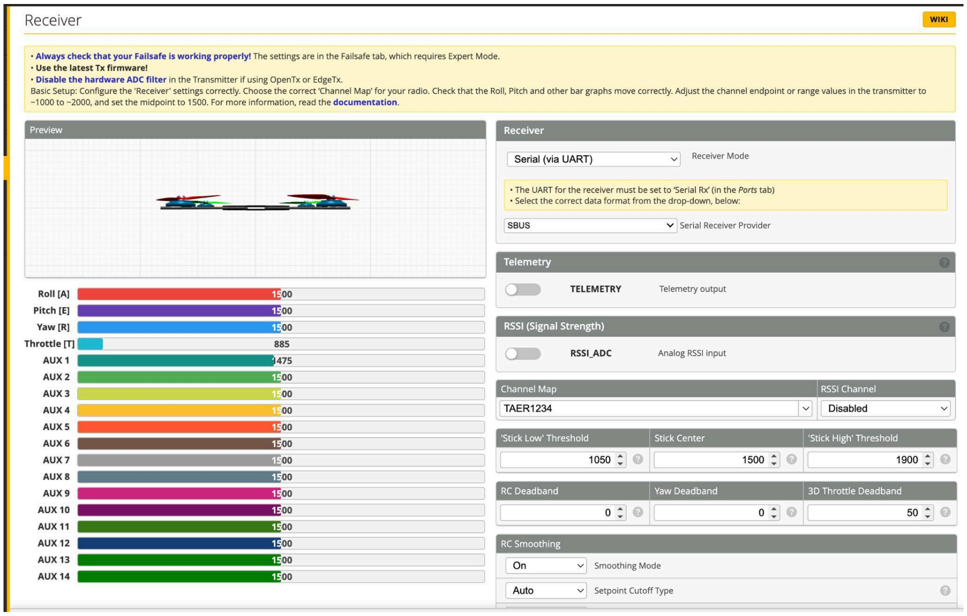


Figure 14: Betaflight Ports Tab

Receiver Tab

Receiver mode is set to Serial (via UART) with SBUS as the serial receiver provider. RC Smoothing is enabled with auto cutoff type. Stick center is 1500, low threshold is 1050, and high threshold is 1900. The 3D Throttle Deadband is set to 50. Channel mapping follows the TAER1234 standard.

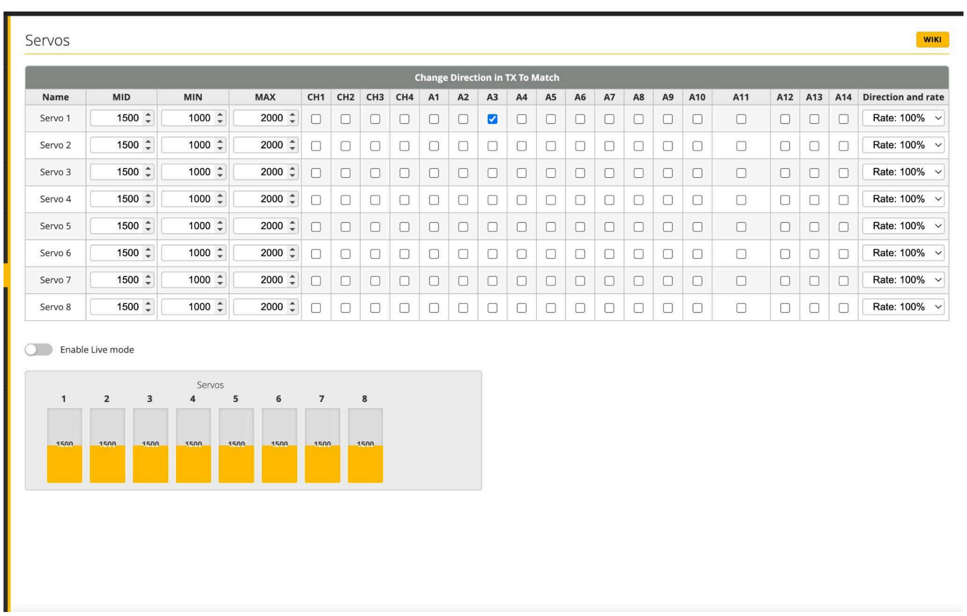


Figure 15: Betaflight Receiver Tab

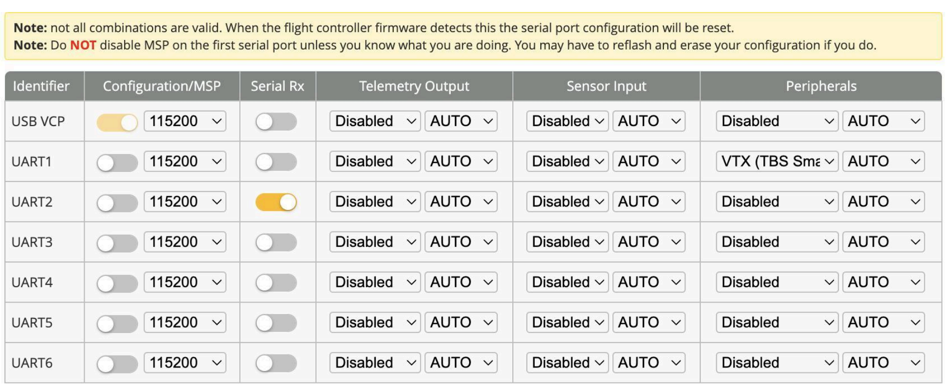
Sensors Tab

The gyroscope graph shows stable readings across all three axes (X, Y, Z) at rest. Gyroscope refresh is set to 50ms with a scale of 2000. The accelerometer is active with a refresh of 50ms at scale 2. Clean gyro traces at rest confirm that the soft-mount setup and RPM filtering are working correctly.

Figure 16: Betaflight Sensors Tab

Servos Tab

Servo 1 is configured on AUX channel A3 to control the claw mechanism. The servo midpoint is 1500, with a range of 1000 to 2000. Direction rate is 100%. All other servo channels remain unconfigured.



Note: not all combinations are valid. When the flight controller firmware detects this the serial port configuration will be reset.
Note: Do **NOT** disable MSP on the first serial port unless you know what you are doing. You may have to reflash and erase your configuration if you do.

Identifier	Configuration/MSP	Serial Rx	Telemetry Output		Sensor Input		Peripherals	
USB VCP	115200		Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART1	115200		Disabled	AUTO	Disabled	AUTO	VTX (TBS Sma	AUTO
UART2	115200		Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART3	115200		Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART4	115200		Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART5	115200		Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART6	115200		Disabled	AUTO	Disabled	AUTO	Disabled	AUTO

Figure 17: Betaflight Servos Tab

Video Transmitter Tab

VTX is configured on Raceband, Channel 1, at 200mW power. The frequency is 5658 MHz. SmartAudio is set as the VTX type. The VTX table includes all standard bands (Boscam A/B/E, Fatshark, Raceband, IMD6) with 8 channels each and 3 power levels (7, 16, 25).

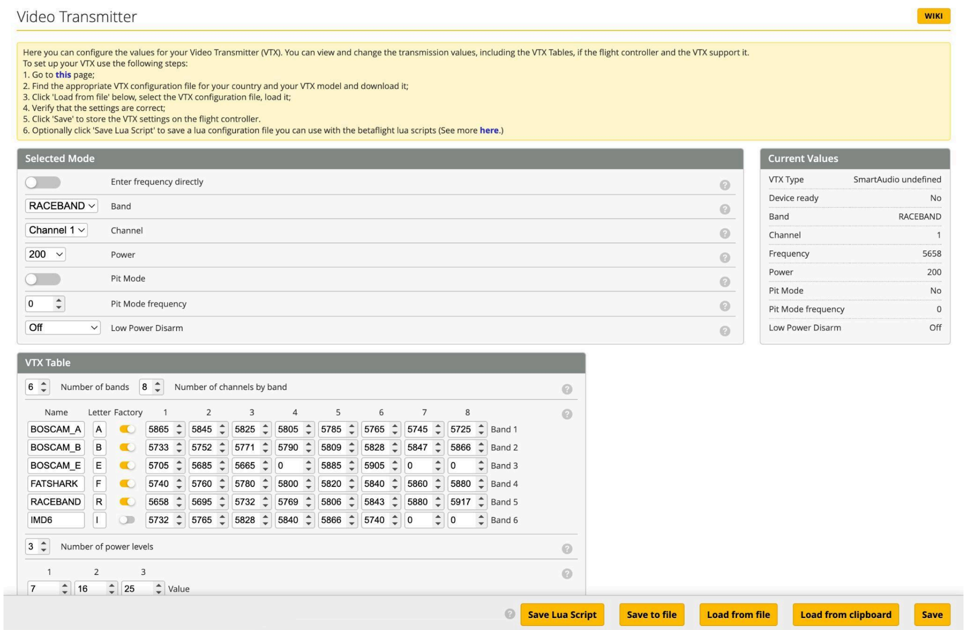


Figure 18: Betaflight Video Transmitter Tab

Power and Battery Tab

Battery monitoring uses onboard ADC for both voltage and current. Minimum cell voltage is set to 3.3V, maximum to 4.3V, with a warning threshold at 3.49V. Capacity is set to 1550 mAh. The voltage meter scale is 110 with a divider value of 10.

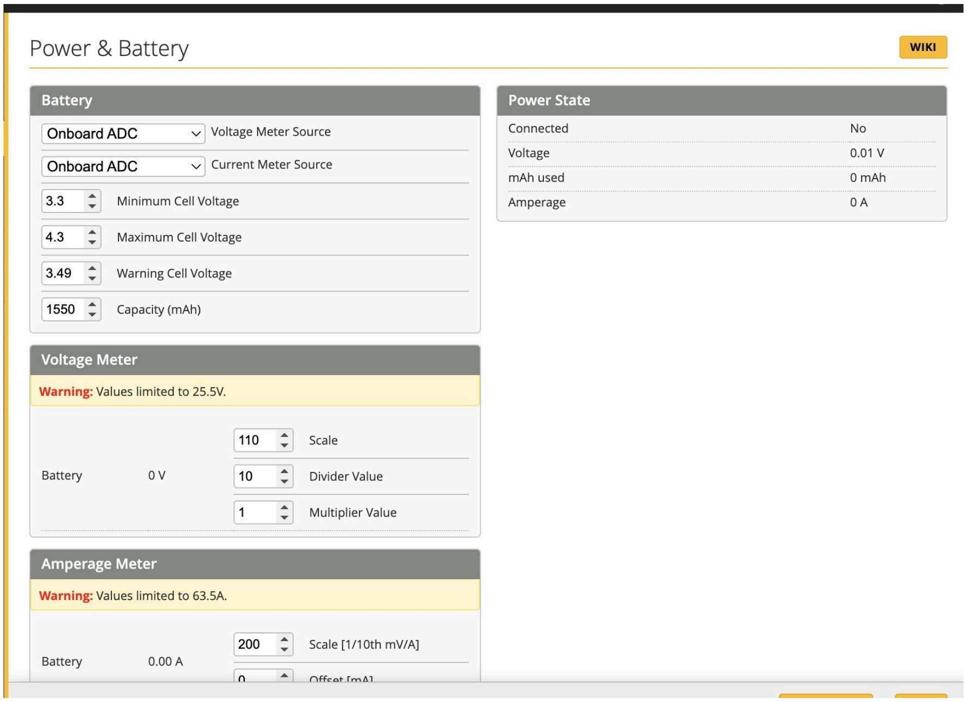


Figure 19: Betaflight Power and Battery Tab

Tuning Process

Blackbox logging was used to log flight data and gyro traces and PID error were analyzed in Betaflight Blackbox Explorer. There were three stages of the tuning process. To begin with, we adjust P-term to a level that allows a responsive stick feel with no visible vibration. Second, we decreased D-term until high-frequency vibration were no longer visible in gyro logs. Third, we tuned I-term to remove drift when making long hovers.

Dynamic notch filtering is implemented using RPM, which measures the harmonics of the motor and removes the noise of vibration before it can be applied to the PID controller. The ESC has the DShot protocol, which is bidirectional. Throttle smoothing is also activated that eliminates the abrupt throttle spikes in performing the precision hovering operation.

Altitude Hold (Planned)

The betaflight 6.0 beta includes horizontal altitude hold, which relies on barometer data to control the plane to a set altitude without pilot intervention. We will incorporate the inclusion of an optical flow sensor and ToF rangefinder to provide indoor position and altitude hold. These sensors will be used to complement the barometer in the GPS denied scenarios such as in the indoor competition arenas.

Practice Course Structure

We constructed a complete size practice course on PVC pipe to replicate the competition. The course frame consists of 1/2-inch PVC pipe and tees and elbow fittings. The cost of the material used in the course was about 54.
The course consists of pass-through gates in different heights, elevated areas where animals will be placed, and a specific landing area. Before the competition, we did obstacle runs every day in 2 weeks, monitoring time and accuracy of claw pickups.
The training of pilots was concentrated on three abilities: using smooth throttle control that would allow maintaining a steady hover, paying attention to the accurate use of the yaw control to keep the course on schedule, and combining roll/pitch control that would help to occupy the narrow corridors. All of the pilots obtained at least 10 hours of stick time prior to competing.

Budget

The table below lists all components and materials purchased for both drones and the practice course.

Component	Qty	Unit Cost	Total
T-Motor F7 HD Stack (FC + ESC)	3	$140.00	$420.00
TBS Source One V5 Frame	2	$30.00	$60.00
EMAX ECO II 2207 Motors	8	$20.99	$167.92
Lumenier N2O 1300mAh 6S 150C LiPo	2	$113.99	$227.98
5-inch Propellers (sets of 4)	4	$5.00	$20.00
XT60 Connectors and Wiring	1	$12.00	$12.00
Servo-Actuated Claw Mechanism	2	$15.00	$30.00
3D Printed Landing Gear (A-frame)	2	$4.00	$8.00
PVC Pipe 1/2 in (40 ft)	1	$39.20	$39.20
PVC Tee Fittings	15	$0.40	$6.00
PVC Elbow Fittings	12	$0.75	$9.00
Propeller Guards (3D printed)	2	$6.00	$12.00
Misc Hardware (screws, standoffs, zip ties)	1	$15.00	$15.00
		Estimated Total	**$1,027.10**

We kept total expenditures under $1,100 by selecting mid-range components, reusing tools already available in the school workshop, and 3D printing custom parts (landing gear and propeller guards) in-house.

Safety Procedures

All team members followed strict safety protocols during building, testing, and competition.

- LiPo batteries are stored in fireproof bags when not in use and charged on a non-flammable surface with a balance charger.
- Propellers are removed during all bench work, firmware flashing, and configuration changes.
- Safety glasses are worn during all flight testing.
- A dedicated spotter monitors each flight and has authority to call for an immediate landing.
- Soldering is performed with ventilation, using lead-free solder, with a fire extinguisher accessible.
- Propeller guards are installed on both drones for competition flights.
- The ARM switch is on AUX 1 and must be deliberately activated. The drone cannot arm accidentally.

Lessons Learned

The most important challenge was the management of vibrations. The pre-test flights demonstrated gyro noise was too high and as a result, the PID was erratic. The problem was solved by soft-mounting the flight controller stack and setting up RPM-based dynamic notch filtering. Fix validation We validated the fix by Blackbox logs which had clean gyro traces in all axes.

The handling of wire was a bigger problem than anticipated. During a test hover on our initial construction, loose signal wires got in touch with a propeller and cut the receiver antenna. We have replaced the wiring harness using shorter leads and have attached all the wires to the frame. There were no other incidents.

Special attention was paid to servo channel mapping with the claw mechanism. We had the first set up wired into the incorrect AUX channel so that when the drone armed, the claw would open. This was solved by connecting the servo to the AUX 3 with a dedicated switch on the transmitter. The Betaflight Modes and Servos tabs are now in sync.

Our second design was the A-frame landing gear design. The initial one had straight PVC legs that were too flexible at landing. The 3D printed A-frame is stiffer, lighter and offers a consistent ground clearance to the claw.

Construction of the two drones was the correct one. In case of motor failure in Drone 1 in practice, we immediately transitioned to Drone 2 and went on training without breaking.